DAYANANDA SAGAR COLLEGE OF ENGINEERING

B.E. in CSE

Scheme of Teaching and Examinations 2022

Department of Computer Science and Engineering

Outcome Based Education (OBE) and Choice Based Credit System (CBCS)

(Effective from the academic year 2023-24)

V SEMESTER

Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching Hours /Week			Examination				Credi
					Theory Lecture	Tutorial	Practical Drawing	Duration in hours	CIE Marks	SEE Marks	Total Marks	
					L	T	P					
1	PCC	22CS51	Automata and Compiler Design	TD: CSE PSB: CSE	3	0	0	03	50	50	100	3
2	IPCC	22CS52	Computer Networks	TD: CSE PSB: CSE	3	0	2	03	50	50	100	4
3	IPCC	22CS53	Artificial Intelligence and Machine Learning	TD: CSE PSB: CSE	3	0	2	03	50	50	100	4
4	PCCL	22CSL54	Mobile Application Development Lab	TD: CSE PSB: CSE	0	0	2	03	50	50	100	1
5	PEC	22CS55X	Professional Elective Course-I	TD: CSE PSB: CSE	3	0	0	03	50	50	100	3
6	AEC	22RM56	Research Methodology and IPR	TD: CSE PSB: ECE	2	2	0	03	50	50	100	3
7	HSMS	22ES57	Environmental Studies	TD: Civil PSB: Civil Engg	2	0	0	02	50	50	100	2
Total					**16**	**2**	**8**		**350**	**350**	**700**	**20**

NOTE:A-Management Course for IT branches, B- Circuit Branches, C-ME Allied Branches (AU, AE, ME), D- CV and CH Branches, E-BT Branch

Professional Elective Course-I

22CS551	ETL and Analytics	22CS553	Digital Image Processing
22CS552	Information Security	22CS554	Digital Finance

PCC: Professional Core Course, **PCCL**: Professional Core Course laboratory, **UHV**: Universal Human Value Course, **MC**: Mandatory Course (Non-credit), **AEC**: Ability Enhancement Course, **SEC**: Skill Enhancement Course, **L:** Lecture, **T**: Tutorial, **P**: Practical **S= SDA**: Skill Development Activity, **CIE**: Continuous Internal Evaluation, **SEE**: Semester End Evaluation. **K :** The letter in the course code indicates common to all the stream of engineering. **PROJ**: Project /Mini Project. **PEC**: Professional Elective Course

Professional Core Course (IPCC): Refers to Professional Core Course Theory Integrated with practicals of the same course. Credit for IPCC can be 04 and its Teaching– Learning hours (L : T : P) can be considered as (3 : 0 : 2) or (2 : 2 : 2). The theory part of the IPCC shall be evaluated both by CIE and SEE. The practical part shall be evaluated by only CIE (no SEE). However, questions from the practical part of IPCC shall be included in the SEE question paper. For more details, the regulation governing the Degree of Bachelor of Engineering /Technology (B.E./B.Tech.) 2022-23

Mini Project - II: Mini Project is a laboratory-oriented/hands on course that will provide a platform to students to enhance their practical knowledge and skills by the development of small systems/applications etc. Based on the ability/abilities of the student/s and recommendations of the mentor, a single discipline or a multidisciplinary Mini-project can be assigned to a group having not more than 4 students. **The group of the mini project should be framed and guide should be allocated at the first week of the fifth semester. The final evaluation of the Mini Project – II will be done at the end of sixth semester**.

MOOCs – Courses are to be selected from - NPTEL/SWAYAM/COURSERA/EDx/UDEMY/UDACITY of duration 8-12 weeks. Can be taken from 5 th sem to 7 th sem and should be related courses of 5-7 semester. The course selected

should be approved by respective HoD and has to be completed before completion of VII semester and the prescribed credits are added in VII semester. Students have to compulsorily submit the certificate after completion of the course to the Department. This is Considered as a head of passing and is mandatory for the award of degree. Those, who do not take-up/complete the requirement, will be declared as failed and the degree will not be awarded.
Professional Elective Courses (PEC): A professional elective (PEC) course is intended to enhance the depth and breadth of educational experience in the Engineering and Technology curriculum. Multidisciplinary courses that are added supplement the latest trend and advanced technology in the selected stream of engineering. Each group will provide an option to select one course. The minimum number of students' strengths for offering a professional elective is 10. However, this conditional shall not be applicable to cases where the admission to the program is less than 10.

Academic Year	**2024-2025**	**L: T: P: S:**	**3:0:0:0**
Course Title	**Automata and Compiler Design**	**Course code**	**22CS51**
Credit Units	**3**	**Course Type**	**PCC**
Exam Hours	**3**	**CIE Marks**	**50**
Total Hours:	**40**	**SEE Marks**	**50**

Course objectives:

1. Introduction to fundamental concepts of Automata Theory, Formal Languages and Compiler Design.
2. Activities carried out in different phases of the compiler.
3. Application of Automata Theory and Formal Languages in the field of compiler design.

Course Outcomes [CO]: At the end of the course, student will be able to:

CO1	Understand the fundamental principles of Automata Theory.
CO2	Analyze and design grammars and automata for various language classes, and evaluate the strengths of different computational models.
CO3	Apply computational models from Automata Theory to practical areas such as compilers.
CO4	Create and develop components of a compiler, including lexical analyzers, parsers, and code generators.

Unit	Contents	Hours	COs
Unit – 1	**Introduction:** Central concepts of automata theory, Chomsky Hierarchy, Deterministic finite automata, Non deterministic finite automata, Conversion of NFA to DFA, Minimization of DFA, Regular expression and regular language. Hands on: Problem solving using the tool-Automaton Simulator	8	**1,2**
Unit - 2	Context Free Grammar, Derivation, Parse trees, Ambiguity, Elimination of ambiguity, Left recursion and Left factoring, Push down automata: Determinism and Non-determinism, Design, The languages of PDA Hands on: Problem solving using the tool-Automaton Simulator	8	**1,2**
Unit – 3	Language Processor, Phases of Compiler, Lexical Analysis: The role of Lexical analyser, Input Buffering, Specification of Tokens and Recognition of Tokens	6	**3,4**
Unit – 4	Syntax Analysis: The role of Parser and types of parsing, Shift Reduce parser and its conflicts, Bottom up Parsing: Compute FIRST and FOLLOW, Simple LR Parsers, LALR and CLR Parsers	10	**3,4**
Unit - 5	Semantic Analysis: Syntax Directed Definition, Syntax Directed Translation, Annotated Parse trees, Evaluation order of SDDs, Intermediate code generation: Variants of Syntax trees, Three address code, Issues in the Code generation	**8**	**4**

Text Books:

Sl.#	Author & Edition	Publisher
1.	John E Hopcroft, Rajeev Motwani, Jeffrey D. Ullman,“ Introduction to Automata Theory, Languages and Computation”, Third Edition .	Pearson.
2.	Alfred V.Aho, Monica S.Lam,Ravi Sethi, Jeffrey D. Ullman, “ Compilers Principles, Techniques and Tools”, Second Edition.	Pearson.

Reference Books:

Sl.#	Author & Edition	Publisher
1.	Elain Rich, "Automata,Computability and complexity", 1st Edition.	Pearson Education,2018
2.	K.L.P Mishra, N Chandrashekaran , 3rd Edition , 'Theory of Computer Science".	PHI,2012.
3.	Peter Linz, "An introduction to Formal Languages and Automata ", 3rd Edition.	Narosa Publishers,1998.

E-Learning(Optional):

1.	**https://youtu.be/vTPylAwDb3o?feature=shared**
2.	**https://youtube.com/playlist?list=PLnGWg_0-mty85HjaM9B-Wf7srZt-kcdyJ&feature=shared**

Self-Study Component(Optional):

Unit	Prerequisite [Studied in syllabus]	Topic (Self-Study)	Reference link
Unit – 1	Deterministic and Non-Deterministic Finite Automata	Applications of Finite Automata to lexical analysis, Tokens, patterns and lexemes.	http://www.ijcset.com/docs/IJCSET 15-06-05-037.pdf
Unit – 3	Context Free Grammar s	Introduction to Turing Machines.	https://www3.cs.stonybrook.edu/~pr amod.ganapathi/doc/theory-of-computation/TuringMachines.pdf
Unit – 4	Intermediate Code Generation	Simple code generator.	1495621768Module30_Content_final.pdf (inflibnet.ac.in)
Unit - 5	Basic Blocks and Flow Graphs	Principal sources of optimization.	https://www.brainkart.com/article/The-Principal-Sources-of-Optimization_8191/ http://www.inf.unibz.it/~artale/Compiler/Lectures/slide12.pdf

Reference Books:

Sl.#	Author & Edition	Publisher
1.	Elaine Rich, Automata, Computability and Complexity.	Pearson education
2.	Michael Sipser: Introduction to the Theory of Computation.	Cengage Learning
3.	John C Martin, Introduction to Languages and The Theory of Computation.	McGraw –Hill Publishing Company Limited

CO – PO, CO-PSO MAPPING

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3										3		
CO2	3	3	3	3	3								3		
CO3	3	3	3	3	3								3		
CO4	3	3	3	3	3				2	2		2	3		

Academic Year	**2024-2025**	**L: T: P: S:**	**3:0:2:0**
Course Title	**Computer Networks**	**Course code**	**22CS52**
Credit Units	**04**	**Course Type**	**IPCC**
Exam Hours	**03**	**CIE Marks**	**50**
Total Hours:	**50 (Theory:40 + Laboratory:10)**	**SEE Marks**	**50**

Course objectives: Course covers the following:

1. Introduction to domain knowledge- OSI Model, building blocks and TCP/IP protocols.
2. Inculcate skills to design the networks
3. Use various open source tools to analyze protocols.

Course Outcomes [CO]: At the end of the course, student will be able to:

CO1	Outline the basic principles of networking and the layered architecture model, focusing on their functions and how they interact in network design and operation
CO2	Analyze and assign Internet Protocol addresses to configure and build subnets
CO3	Illustrate how routing algorithms and protocols function in network operations
CO4	Assess how application layer and transport layer protocols influence network design.
CO5	Apply and evaluate error detection and correction techniques to ensure data integrity and reliability in various scenarios.

Unit	Contents	Hours	COs
Unit – 1	**INTRODUCTION :** Overview of the Internet– Network Types, Switching- Circuit switching, Packet switching, The Internet, Accessing the Internet, Protocol Layering: Scenarios, TCP/IP Protocol Suite, Functions of each layer, The OSI Model, Internet History : Early History Birth of the Internet, Internet Today, Standards and Administration **Hands-on:** Demonstration of Cisco Packet Tracer, Simple LAN, Simple Client server	**8** **2**	**CO1**
Unit - 2	**NETWORK LAYER**: Introduction: Network-Layer Services, Structure of A Router Network Layer Protocols: IPv4, Datagram Format, IPv4 Addresses, Classful addressing, Classless addressing , Subnetting, IP Address space management: DHCP – NAT ICMPv4, Unicast Routing , General Idea, Routing Algorithms **Hands-on:** IP Addressing, Sub netting, DHCP, NAT	**8** **2**	**CO1 & CO2**
Unit – 3	**NETWORK LAYER(Contd) :** Distance Vector Routing, RIP, Link State Routing, OSPF, **APPLICATION LAYER** : **Introduction**: Providing Services, **Application-Layer Paradigms:** Client-Server , Peer-to-peer, **Application Programming Interface,** Sockets, **Using Services of the Transport Layer** , **Standard Client-Server applications**: World Wide Web and HTTP, **Domain Name System (DNS),** Need, Hierarchy, DNS Resolution, additional services **Hands-on:** Static Routing, Dynamic Routing, DNS	**8** **2**	**CO3 & CO4**
Unit – 4	**TRANSPORT LAYER** – Introduction, Transport-Layer Services **Transmission Control Protocol (TCP)** : TCP Services, Features, Segment, Connection management, Windows in TCP, Flow Control, Error Control, Congestion Control, Timers, **User Datagram Protocol** **Hands-on:** VOIP	**8** **2**	**CO1 & CO4**
Unit - 5	**DATA LINK LAYER** : Introduction, Data link services, Error detection – Cyclic Redundancy Check, Link layer addressing, ARP, **Wired LAN**: **Building blocks:** Connecting devices : Repeaters or Hubs, Link-Layer Switches, Self-learning	**8**	

	principle of LAN Switch, Routers, **Virtual LANs-** Need, different types of VLANs. **Hands-on:** VLan, CRC	**2**	**CO5**

Text Books:

Sl.#	Author & Edition	Publisher
1.	Behrouz A Forouzan, Firouz Mosharraf (2012)- Computer Networks (A Top-down Approach)- Special Indian Edition, Pearson	Pearson, 2012.
2.	James F Kurose, Keith W (2012), Computer Networking (A Top-down approach),6th Edition, Pearson	Pearson, 2012.

Reference Books:

Sl.#	Author & Edition	Publisher
1.	Computer Networks: A Systems Approach- Larry Peterson – 5th Edition – MK	Larry Peterson

E-Learning:

1.	**https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013843377125924864369 33_shared/overview**
2.	https://youtube.com/playlist?list=PLDQaRcbiSnqF5U8ffMgZzS7fq1rHUI3Q8&si=OauQJZ_WJ5kImcwB

Self-Study Component:

- Telephony network architecture
- Characteristics of different information types : Voice, Text & Video
- IPV6- Next Generation Internet Protocol
- Network utility tools : PING, Traceroute, ipconfig, Nslookup etc
- Border Gateway Protocol
- Email architecture
- Reliable Data transfer protocols : Stop-n-wait, Go back N & Selective repeat
- Media Access Control protocols fundamentals – CSMA/**CD**

Course outcomes

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	**2**	**2**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**3**
CO2	**2**	**3**	**2**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**3**
CO3	**2**	**2**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**3**
CO4	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**3**
CO5	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**3**

Academic Year	**2024-2025**	**L: T: P: S:**	**3:0:2:0**
Course Title	**ARTIFICIAL INTELLIGENCE & MACHINE LEARNING**	**Course code**	**22CS53**
Credit Units	**04**	**Course Type**	**Integrated**
Exam Hours	**03**	**CIE Marks**	**50**
Total Hours:	**50 [T:Theory 40 + L:Laboratory 10]**	**SEE Marks**	**50**

Course objectives: Course will cover the following:

1. The fundamentals of Artificial Intelligence, Intelligent Agents and knowledge representation
2. Different supervised and unsupervised machine learning algorithms
3. Clustering techniques for organizing and retrieving data
4. Introduction to deep nets and their applications

Course Outcomes [CO]: At the end of the course, students will be able to:

CO1	Comprehend the working of intelligent agents and their working environments.
CO2	Represent knowledge and use it for reasoning.
CO3	Apply linear regression to build machine learning models to solve real life problems.
CO4	Analyze and apply supervised techniques for classification.
CO5	Evaluate and select suitable clustering technique for problem solving.
CO6	Design and implement solutions using machine learning techniques to solve complex problems.

Unit	Contents	Hours	COs
Unit– 1	**Topics:** Introduction to Artificial Intelligence, Intelligent Agents: Intelligent Agents: Agents and environment; The nature of environment; The structure of agents. Logical Agents: Logical Agents,: Knowledge-Based Agents, The Wumpus World, Logic, Propositional logic. Difference between Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL), Supervised, Unsupervised Learning semi-supervised learning. Reinforcement Learning, Regression vs Classification problems.	08	**CO1,**
	Hands-on: Learning Python using the following resources: **i) Python for Beginners: https://www.w3schools.com/python/** **ii) Python for Data science: https://www.coursera.org/learn/python-for-applied-data-science-ai#syllabus** **iii) https://www.learndatasci.com/tutorials/reinforcement-q-learning-scratch-python-openai-gym/**	02	**CO6**
Unit - 2	**Topics:** First-Order Logic: Using first-order logic. Inference and Reasoning: Inference in First-Order Logic. Unification and lifting, Forward chaining; Backward chaining; Resolution.	08	**CO2,**
	Hands-on: Experimenting Inferencing techniques using python libraries	02	**CO6**

Unit – 3	**Topics:** Supervised learning: Linear Regression & Classification: Linear Regression: Simple Linear Regression: Estimating the Coefficients, Assessing the Accuracy of the Coefficient Estimates, Assessing the Accuracy of the Model Multiple Linear Regression: Estimating the Regression Coefficients Classification: Basic Concepts, Decision Tree Induction, Bayes Classification Methods, Model Based Evaluation and Selection: Metrics for Evaluating Classifier Performance, Comparing Classifiers Based on Cost–Benefit and ROC Curves	08	**CO3, CO4**
	Hands-on: Design and implement solutions using Linear Regression, Naive bayes, decision tree, SVM models, Association Rules	02	**CO6**
Unit – 4	**Topics:** Unsupervised Learning Techniques: Clustering Clustering: Partitioning Methods: K-means; Hierarchical Methods: Agglomerative versus Divisive Hierarchical Clustering, Distance Measures in Algorithmic Methods, BIRCH, Density-Based: DBSCAN Evaluation of Clustering: Assessing Clustering Tendency, Determining the Number of Clusters, Measuring Clustering Quality.	08	**CO5**
	Hands-on: Design and implement solutions using K-Means and Hierarchical clustering approaches	02	**CO6**
Unit - 5	**Topics:** Neural Networks and Deep Learning: Feed forward Neural Networks: Unit, Expressing the Neural Network in Vector Form, Evaluating the output of the Neural Network. Types of Units/Activation Functions/Layers Deep Learning: Convolutional neural networks (CNN), Recursive (Recurrent) Neural Networks (RNN).	08	**CO6**
	Hands-on: Design and implement solutions using Neural Networks, CNN and RNN in Tensorflow/PyTorch	02	

Text Books:

Sl.#	Author & Edition	Publisher
1.	Stuart Russel, Peter Norvig: Artificial Intelligence A Modern Approach	3rd Edition, Pearson Education, 2010.
2.	Data Mining Concepts and Techniques", Jiawei Han, Micheline Kamber, Jian Pei	Elsevier (MK) 3rd Edition, 2012.
3.	Deep Learning with Python: A Hands-on Introduction Nikhil Ketkar	
4.	An Introduction to Statistical Learning, with Applications in R (1913), by G.James, D. Witten, T. Hastie, and R. Tibshirani.	
5.	https://towardsdatascience.com/notes-on-artificial-intelligence-ai-machine-learning-ml-and-deep-learning-dl-for-56e51a1971c2	

Reference Books:

Sl.#	Author & Edition	Publisher
1.	Tom M. Mitchell, "Machine Learning",	McGraw-Hill Education (INDIAN EDITION), 1997.
3.	Nils J. Nilsson: Principles of Artificial Intelligence	Elsevier, 1980.

E-Learning:

1.	https://youtu.be/jUE3ciHrcxM
2.	https://youtu.be/-5hEYRt8JE0
3.	https://www.youtube.com/watch?v=SGZ6BttHMPw

Self-Study Component:

Unit	Prerequisite [Studied in syllabus]	Topic (Self-Study)	Reference link
Unit – 1	AI	What is AI? How to become a practitioner with machine learning? AI & ethics	https://www.youtube.com/watch?v=NbEbs6I3eLw
Unit – 2	Propositional Logic	Syntax and semantics of first-order logic Propositional versus first-order inference	https://www.youtube.com/watch?v=itrXYg41-V0
Unit – 3	Classification	Data preparation/cleaning Techniques to improve classification accuracy Bayesian Belief networks Support Vector Machines	https://www.youtube.com/watch?v=xG-E--Ak5jg
Unit – 4	Unsupervised learning	Basic Concepts and Methods, Cluster Analysis overview Association rules (Recommendation Systems)	https://www.youtube.com/watch?v=D6gtZrsYi6c
Unit - 5	Artificial Neural Networks	Gradient descent MLOps	https://www.youtube.com/watch?v=aircAruvnKk

CO – PO, CO-PSO MAPPING

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3						1	1		3	3		
CO2	3	3	3						1	1		3	3		
CO3	3	3	3		3	2			2	2		3	3		
CO4	3	3	3		3	2			2	2		3	3		
CO5	3	3	3		3	2			2	2		3	3		
CO6	3	3	3		3	2			2	2		3	3		

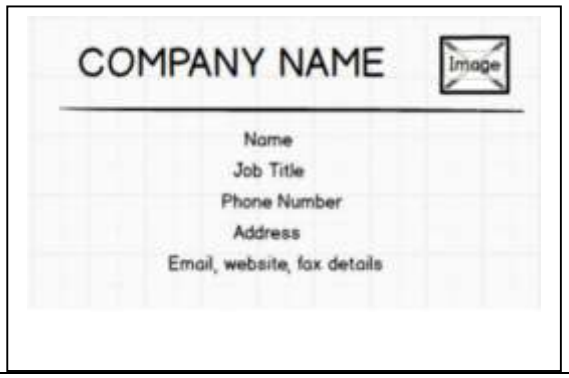
Academic Year	**2024-2025**	**L: T: P: S:**	**0:2:0:0**
Course Title	**Mobile Application Development Lab**	**Course code**	**22CSL54**
Credit Units	**1**	**Course Type**	**PCCL**
Exam Hours	**3 hours**	**CIE Marks**	**50**
Total Hours:	**2 hours**	**SEE Marks**	**50**

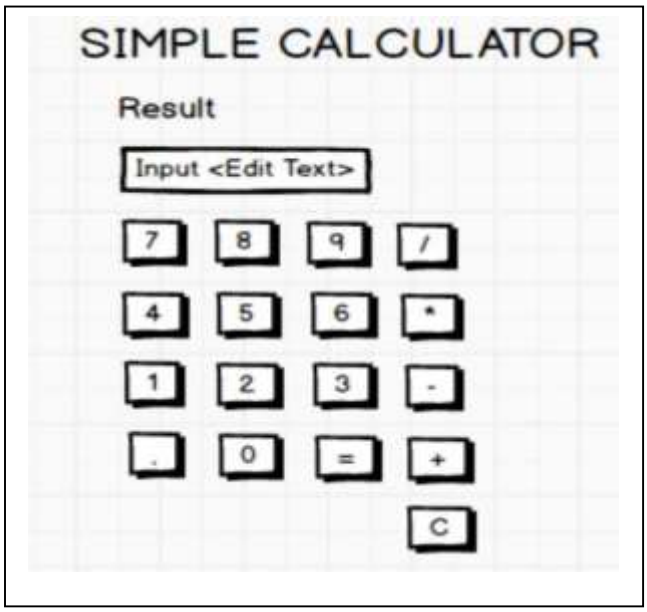
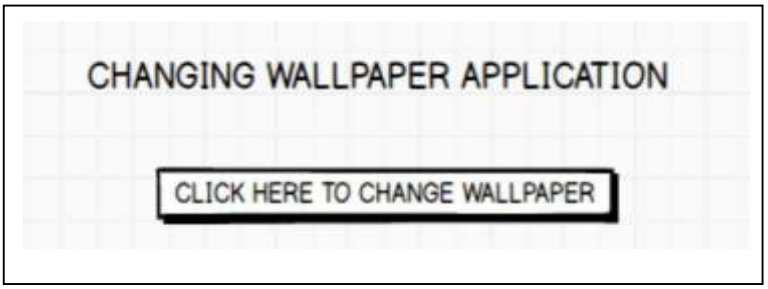
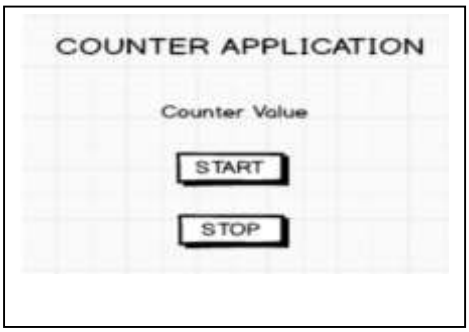
Course objectives:

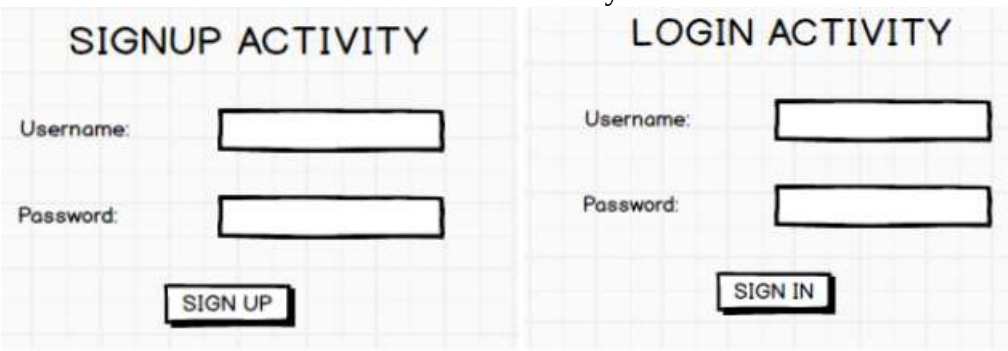
1. Learn and acquire the art of Android Programming.
2. Configure Android studio to run the applications.
3. Understand and implement Android's User interface functions.
4. Create, modify, and query on SQLite database.
5. Inspect different ways to accomplish background processing in Android

Course Outcomes: At the end of the course, students will be able to:

CO1	Develop, test and debug android applications in Android Studio.
CO2	Design adaptive, responsive user interfaces with backend data processing that work across a wide range of mobile devices.
CO3	Develop form based applications for simple tasks using data storage and retrieval libraries.

Expt No	**Content of the Experiment**	**Hours**	**CO's**
1.	Introduction to Mobile Application Development, Android Studio installation and project creation. Create an application to design a **Dream Job Visiting Card**. The Visiting card should have a company logo at the top right corner. The company name should be displayed in Capital letters, aligned to the center. Information like the name of the employee, job title, phone number, address, email, fax and the website address is to be displayed. Insert a horizontal line between the job title and the phone number . 	**2**	**CO 1**

2.	Develop an Android application using controls like Button, TextView, EditText for designing a **Calculator** having basic functionality like Addition, Subtraction, Multiplication, and Division.  	**2**	**CO 1**
3.	Develop an application to set an image as **wallpaper**. On click of a button, the wallpaper image should start to change randomly every 30 seconds.  	**1**	**CO 1**
4.	Write a program to create an activity with two buttons START and STOP. On Pressing of the START button, the activity must start the **counter** by displaying the numbers from One and the counter must keep on counting until the STOP button is pressed. Display the counter value in a TextView control.  	**1**	**CO2**
5.	Create a SIGN-UP activity with Username and Password. • **Password should contain uppercase and lowercase letters.** • **Password should contain letters and numbers.** • **Password should contain special characters.** • **Minimum length of the password (the default value is 8).**	**2**	**CO2**

	Password should be validated based on rules. On successful **SIGN UP** proceed to the next Login activity. Here the user should **SIGN IN** using the Username and Password created during signup activity. If the Username and Password are matched, then navigate to the next activity which displays a message saying "Successful Login" or else display a toast message saying "Login Failed". The user is given only two attempts and after that display a toast message saying "Failed Login Attempts" and disable the SIGN IN button. Use Bundle to transfer information from one activity to another. 		
6.	Write a program to enter **Medicine** Name, Date and Time of the Day as input from the user and store it in the SQLite database. Input for Time of the Day should be either Morning or Afternoon or Evening or Night. **Trigger an alarm** based on the Date and Time of the Day and display the Medicine Name.	**2**	**CO 2**
7.	Mini Project- Develop an Android Application of your choice which implements all the concepts covered. Demonstration with Report should be submitted.	**2**	**CO3**

Reference Books:

1. Erik Hellman, "Android Programming – Pushing the Limits", 1st Edition, Wiley India Pvt Ltd, 2014. ISBN-13: 978-8126547197.
2. Dawn Griffiths and David Griffiths, "Head First Android Development", 1st Edition, O"Reilly SPD Publishers, 2015. ISBN-13: 978-9352131341.
3. Bill Phillips, Chris Stewart and Kristin Marsicano, "Android Programming: The Big Nerd Ranch Guide", 3rd Edition, Big Nerd Ranch Guides, 2017. ISBN-13: 978-0134706054

CO-PO, PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO 12	PSO1	PSO2	PSO3
CO1	3	3	3	-	2	-	-	1	2	1	-	3	3	-	-
CO2	3	-	3	-	2	-	-	1	2	1	-	1	3	-	-
CO3	3	3	3	-	2	-	-	1	2	1	-	3	3	-	-

Academic Year	**2024-2025**	**L: T: P: S:**	**0:3:0:0**
Course Title	**ETL and Analytics**	**Course code**	**22CS551**
Credit Units	**3**	**Course Type**	**PEC**
Exam Hours	**3**	**CIE Marks**	**50**
Total Hours:	**40**	**SEE Marks**	**50**

Course objectives:

1. To introduce the fundamental processes of ETL and Data warehousing
2. To impart the knowledge on various data analytics concepts and techniques that can be applied to Data science, Data engineering
3. To develop knowledge of analytical skills which shall be able to solve real time problems

Course Outcomes [CO]: At the end of the course, student will be able to:

CO1	Apply the skills of extracting data from various sources and loading them to data marts and data warehouses
CO2	Cleaning preprocessing and transforming of data using sql for building ML and AI models
CO3	Demonstrate capacity to perform a self-directed piece of practical work that requires the application of analytical techniques.
CO4	Understand the need of data warehousing and data lake architecture

Unit	Contents	Hours	COs
Unit – 1	**Introduction to ETL —**OLAP vs OLTP , concept of extract transform and loading, common techniques used , introduction to data warehousing and data marts, using sql for transformations **Revisit SQL:** **Application of Joins** – Natural joins , Cross joins, inner join, left joins, outer joins, complex transformations using joins , self joins **Sub queries and correlated sub queries -** using EXISTS , Row valued function , common table expressions **Case study** – using MYSQL to perform transformations on banking and hr databases	8	CO2,CO3
Unit - 2	**Advanced SQL for Analytics :** **Window functions –** Rank functions, Ntile, lead, lag, last_value, first_value, aggregation functions **Few Advanced concepts -** Recursion in SQL , pivot function in SQL, views in SQL **Case study** – using MYSQL to perform transformations on banking and hr databases	8	CO2,CO3
Unit – 3	**Introduction to Data warehousing -**Kimball approach to data warehousing, fact tables, dimension tables, slowly changing dimensions type 1, type 2 , type 3 and type 4,upsert operations, soft delete **Case study** – using MYSQL to perform upsert operations and SCD type 1 and type 2	8	CO1, CO4
Unit – 4	**Introduction to Data lake architecture** – processing semi structured data like json, compression techniques and file formats used for big data, Case study - using data bricks for performing ETL	8	CO1, CO4

Unit - 5	**Introduction to Data Integration tools –** practical hands on using python to fetch data from different sources, using Pentaho for ETL , using apache kafka and apache spark for transformation	8	CO4

Text Books:

Sl.#	Author & Edition	Publisher
1.	Learning SQL: Generate, Manipulate, and Retrieve Data, Third Edition ()	Greyscale Indian Edition
2.	The Datawarehousing tool kit- Kimball , Third Edition,	Wiley publications

Reference Books:

Sl.#	Author & Edition	Publisher
1.	**SQL All - In - One For Dummies, 3ed**	
2.	Understanding ETL	O'Reilly publications
3.	Apache Spark – Definitive guide	O'Reilly publications

E-Learning(Optional):

1.	**Pentaho Data Integration -Udemy** udemy link for pentaho
2.	**udemy link for data lake**

Academic Year	**2024-2025**	**L: T: P: S:**	**3:0:0**
Course Title	**Information Security**	**Course code**	**22CS552**
Credit Units	**3**	**Course Type**	**PEC-I**
Exam Hours	**3**	**CIE Marks**	**50**
Total Hours:	**40**	**SEE Marks**	**50**

Course objectives:

1. To provide an understanding of principal concepts, major issues, technologies, and basic approaches in information security.
2. Master the key concepts of information security and how they "work."
3. Develop a "security mindset:" learn how to critically analyze situations of computer and network usage from a security perspective, identifying the salient issues, viewpoints, and trade-offs.

Course Outcomes [CO]: At the end of the course, student will be able to:

CO1	Evaluate vulnerability of an information system and establish a plan for risk management.
CO2	Demonstrate basic principles of Web application security
CO3	Evaluate the authentication and encryption needs of an information system.
CO4	Demonstrate how to secure a network

Unit	Contents	Iours	COs
Unit – 1	**Computer Security and Technology principles:** Computer security concepts, Threats, Attacks, Assets, Security functional requirement, fundamental security design principle, attack surface, attack trees, Computer security strategy. Cryptographic tools, User Authentication, Access control, Database and cloud security, Malicious software, Denial of Service attacks, Intrusion detection, Firewalls and Intrusion prevention system.	8	CO1
Unit - 2	**Software Security and Trusted Systems**: **OS security**-introduction, system security planning, OS hardening, application security, security maintenance, Trusted Computing and multilevel security- Different models of computer security, Concept of trusted systems, application of multilevel security. **Cloud and IOT**	8	CO2
Unit – 3	**Cryptographic Algorithms**: Symmetric Encryption and message confidentiality- symmetric encryption principles, DES, AES, Stream ciphers and PC4, Cipher Block Modes of operations,	8	CO3
Unit – 4	**Public key Cryptography and Message Authentication**- Secure hash algorithm, HMAC, RSA, Diffie-hellman.	8	CO4
Unit - 5	**Internet Security:** Internet Security protocol and standards-secure Email, S/MIME, SSL, TLS, HTTPS,IPv4vs IPv6 security Internet Authentication Applications- kerberos, C.509, Public key infrastructure	8	CO4

<u>Text Books:</u>

Sl.#	Author & Edition	Publisher
1	Computer Security: Principles and Practice, William Stallings; Lawrie Brown	

Reference Books:

Sl.#	Author & Edition	Publisher
1.	Introduction to Computer Security, 2004 Matt Bishop,	Addison-Wesley, ISBN 0-321-24744-
2.	Buchmann J. A., Introduction to Cryptography, Springer Verlag (2001).	
3	Stallings William, Cryptography and Network Security	
4	Schneier Bruce, Applied Cryptography, John Wiley and Sons (1996).	
5	Britz M., Computer Forensic and cyber crime, Upper Saddle River, Prentice Hall (2003).	

Academic Year	**2024-2025**	**L: T: P: S:**	**3:0:0:0**
Course Title	**Digital image processing**	**Course code**	**22CS553**
Credit Units	**3**	**Course Type**	**PEC**
Exam Hours	**3**	**CIE Marks**	**50**
Total Hours:	**40**	**SEE Marks**	**50**

Course objectives:

1. To understand the fundamental concepts of image processing.
2. To learn various image processing techniques and their applications.
3. To gain practical experience in implementing image processing algorithms.

Course Outcomes [CO]: At the end of the course, student will be able to:

CO1	Explain the fundamental concepts and principles of digital image processing.
CO2	Implement image enhancement and restoration techniques using programming tools such as MATLAB or Python.
CO3	Design and implement algorithms for image segmentation and object recognition
CO4	Analyze and solve real-world problems using digital image processing techniques.

Unit	Contents	Hours	COs
Unit – 1	**Fundamentals of Image Processing: Introduction to Digital Image Processing** Definition and types of digital images, Image formation and acquisition, Sampling and quantization **Mathematical Foundations:** Linear algebra and matrix operations, Convolution and correlation, Fourier Transform	6	
Unit - 2	**Image Enhancement Techniques: Spatial Domain Methods**-Point processing (contrast stretching, histogram equalization) Spatial filtering (smoothing, sharpening) **Frequency Domain Methods**: Fourier Transform and properties, Frequency domain filtering (low-pass, high-pass filters) **Hands-On :** Implementing enhancement techniques using MATLAB/Python	9	
Unit – 3	**Image Restoration Techniques**: **Noise Models**-Types of noise (Gaussian, salt-and-pepper), Noise reduction techniques **Restoration Algorithms**: Inverse filtering, Wiener filtering, Adaptive filtering **Case Studies** Image restoration applications in medical imaging and photography	9	
Unit – 4	**Image Segmentation and Recognition: Segmentation Techniques**-Thresholding (global and adaptive),Edge detection (Sobel, Canny),Region-based segmentation (region growing, region splitting) **Object Recognition**-Feature extraction (texture, shape, color), Pattern recognition algorithms (template matching, machine learning techniques) • **Projects and Presentations:** Developing segmentation and recognition algorithms for real-world applications	**7**	
Unit - 5	**Image Compression**: Lossless compression (Huffman coding, run-length encoding), Lossy compression (JPEG, JPEG2000),Wavelet-based compression **Color Image Processing**: Color models (RGB, CMY(K), HSI, YCbCr),Color enhancement and transformation	**9**	

	Applications and Case Studies • Medical image analysis • Autonomous vehicles • Surveillance systems • Industrial inspection		

Text Books:

Sl.#	Author & Edition	Publisher
1.	"Digital Image Processing" by Rafael C. Gonzalez and Richard E. Woods	Pearson

Reference Books:

Sl.#	Author & Edition	Publisher
1.	"Fundamentals of Digital Image Processing" by Anil K. Jain	Prentice Hall
2.	"Image Processing, Analysis, and Machine Vision" by Milan Sonka, Vaclav Hlavac, and Roger Boyle	Cengage Learning

Academic Year	**2024-2025**	**L: T: P: S:**	**3:0:0:0**
Course Title	**Digital Finance**	**Course code**	**22CS554**
Credit Units	**3**	**Course Type**	**PEC-1**
Exam Hours	**3**	**CIE Marks**	**50**
Total Hours:	**40**	**SEE Marks**	**100**

Course objectives:

1. Understand the fundamentals of digital finance.
2. Understanding the key notions and concept shift from traditional to digital finance.
3. Analyze the impact of blockchain and cryptocurrencies on the financial industry.
4. Evaluate digital payment systems and their applications.
5. Understand emerging technologies, and the regulatory environment of digital finance.

Course Outcomes [CO]: At the end of the course, student will be able to:

CO1	An in-depth exploration of the digital finance landscape.
CO2	Analyzing Security and fraud prevention in digital payments.
CO3	Interpret the Potential of Security Tokens and Digital Assets.
CO4	Infer various bitcoin related security and privacy issues and future opportunities in digital finance.
CO5	Analyzing the various digital finance tools and risks involved in digital payments.

Unit	Contents	Hours	COs
Unit – 1	**The Digital Financial Ecosystem and Its Components.** The Evolution of the DFS Ecosystem. CICO Problem and payment Services. **The Magic Ledger** What Is Blockchain? Foundational Elements of Blockchain. Distributed Ledger Technology. Building a Blockchain. Creating a Blockchain Transaction. Tampering with a Blockchain. Smart Contracts. Oracles. The Characteristics and Benefits of Blockchain What Differentiates One Blockchain Platform from Another?	8	**CO1**
Unit - 2	**Fundamentals Of a Security Token:** Distinguishing the Types of Digital Assets. How Security Tokens Came Along. Tokenizing a Security. Benefits of Tokenization. Fundamentals of a Security Token, What Types of Assets May Be Tokenized? Key Features of a Security Token, The Security Token Ecosystem.	8	**CO2**
Unit – 3	**Markets for Digital Assets & Security Tokens.** Trading & Finding Liquidity. Developing the Primary Markets. Bringing the Token to Market. Developing Secondary Markets. What Is Needed for Tokenization to Thrive? Case Study: Ripple, Case Study: Singapore's iSTOX Digital Platform	8	**CO3**
Unit – 4	**"DeFi": eLending and the Future of Getting a Loan**. Crypto Lending & Borrowing Earn Interest on Crypto. Making Loans with Blockchain. Mortgages. Security and fraud prevention in digital payments, Case Study: Machine learning in stock market predictions. Case Study: The impact of artificial intelligence and machine learning. Case Study: Future challenges and opportunities in digital finance.	8	**CO4**
Unit - 5	**Digital Payments and Cyber Risk** Overview of popular digital finance tools and platforms, Types of digital payment systems (e.g., mobile payments, e-wallets, online banking) **Risks** Smart Contract Risk Governance Risk Oracle Risk Scaling Risk DEX Risk Custodial Risk Regulatory Risk.	8	**CO5**

Text Books:

Sl.#	Author & Edition	Publisher
1.	**Digital Finance:** Security Tokens and Unlocking the Real Potential of Blockchain Baxter Hines	Wiley
2.	**DeFi and the Future of Finance** Ashwin Ramachandran, Fred Ehrsam (Foreword	Wiley

	by), Vitalik Buterin (Preface by), Joey Santoro, Campbell R. Harvey.	

Reference Books:

Sl.#	Author & Edition	Publisher
1.	"The Age of Cryptocurrency: How Bitcoin and Digital Money Are Challenging the Global Economic Order" by Paul Vigna and Michael J. Casey	St. Martin's Press
2.	**Digital Finance, Big Data, Start-ups, and the Future of Financial Services By Perry Beaumont**	Routledge
3.	Decentralizing Finance: How DeFi, Digital Assets, and Distributed Ledger Technology Are Transforming Finance Kenneth Bok.	Wiley

E-Learning(Optional):

1.	**Udemy course in Digital finance**
2.	**https://nptel.ac.in/courses/106106139**

CO – PO, CO-PSO MAPPING

Course	PO1	PO2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO1	PSO2
CO1	2	2	2	1	2	2	-	-	-	-	1	3	1	1
CO2	2	2	2	1	2	2	-	-	-	-	1	3	1	1
CO3	2	2	2	1	2	2	-	-	2	-	1	3	1	1
CO4	2	3	2	2	2	2	-	-	3	3	1	3	2	2
CO5	2	2	2	1	3	2	-	-	2	-	1	3	1	1

Academic Year	**2024-2025**	**L: T: P: S:**	**2:1:0:0**
Course Title	**Research Methodology and IPR**	**Course code**	**22RM56**
Credit Units	**3**	**Course Type**	**Elective**
Exam Hours	**3**	**CIE Marks**	**50**
Total Hours:	**40**	**SEE Marks**	**50**

Course Objectives:

1	To identify the importance of research.
2	To understand the research methodology, tools and techniques.
3	To interpret the results of Research and Propose recommendations.
4	To understand Intellectual Property rights.

Course Outcomes:

After completion of the course, the graduates will be able to …

CO1	Outline the importance of conducting research, and Intellectual property rights
CO2	Analyze different types of research problem and criteria for good research.
CO3	Examine features and importance of research design including collection and sampling of data
CO4	Develop Intellectual property rights documents and technical reports.

Module	Course Contents	Hours	COs
1	Meaning, Objectives and Characteristics of research - Research methods Vs Methodology - Types of research -Descriptive Vs. Analytical, Applied Vs. Fundamental, Quantitative Vs. Qualitative, Conceptual Vs. Empirical - Research process - Criteria of good research – Developing a research plan, Case Study and Practice Session.	8	CO1, CO2, CO3
2	**Defining the research problem**- Selecting the problem- Necessity of defining the problem - Techniques involved in defining the problem- Importance of literature review in defining a problem-Survey of literature-Primary and secondary sources-Reviews, treatise, monographs - patents-web and source- searching the web-Identifying gap areas from literature review-Development of working hypothesis, Case Study and Practice Session.	8	CO2, CO3
3	**Research design and methods**–Research design–Basic Principles-Need of research design– Features of good design, Development of Models-Developing a research plan-exploration, Description, Diagnosis, and Experimentation – Determining experimental and sample designs, **Sampling design-**Steps in sampling design-Characteristics of good sample design-Types of sample designs- Measurement and scaling techniques-**Methods of data collection**-Collection of primary data- Data collection instruments Testing of hypotheses-Basic Concepts-Procedure for hypotheses testing flow diagram for hypotheses testing. Case Study and Practice Session.	8	CO1, CO2, CO3
4	**Interpretation and report writing-** Techniques of interpretation- Structure and components of scientific reports- Different steps in the preparation – Layout, structure and language of the report - Illustrations and tables- Types of report-Technical reports and thesis, Case Study and Practice Session **IPRs**- Role of IP in the Economic and Cultural Development of the Society, IP Governance, IP as a Global Indicator of Innovation, Invention and Creativity –Intellectual Property-Importance and Protection of Intellectual Property Rights (IPRs), Case Study and Practice Session.	8	CO1, CO4
5	**A brief summary of**: Patents, Copyrights, Trademarks, Industrial Designs-Integrated Circuits – Geographical Indications — Application and	8	CO1, CO4

	procedures. **World Intellectual Property Organization** (WIPO): Introduction, Origin of WIPO, Salient Features of WIPO, Main Activities of WIPO, India and WIPO, Access to International IP Filing Systems, Protection of Traditional Knowledge (TK), Indian web portals for patents & technologies, Case Study and Practice Session.		

Note:

1. Questions for CIE and SEE not to be set from self-study component.
2. Assignment Questions should be from self-study component only.

Self-Study Components:

Module 1: Understand the combination of both qualitative and quantitative methods.

Module 2: Learn how to identify and articulate research problems.

Module 3: Learn how to conduct a thorough literature review to support your research.

Module 4: Study how to develop testable hypotheses.

Module 5: Parliamentary Standing Committee Report on IPR Regime and Addressing IP Crimes.

Text Books:

1	“Research Methodology: Methods and Techniques” (4th Edition, 2019) by Kothari, C.R., New Age International.
2	“Intellectual Property A Primer for Academia” (1st Edition, 2021) by Prof. Rupinder Tewari Ms. Mamta Bhardwaj.
3	"Research Design: Qualitative, Quantitative, and Mixed Methods Approaches" (6th Edition, 2023) by John W. Creswell and J. David Creswell
4	"Research Methodology: A Step-by-Step Guide for Beginners" (4th Edition, 2023) by Ranjit Kumar
5	"Intellectual Property Rights in an Age of Electronics and Information" (2023) by John V. Martin

Reference Books:

1	"Statistics with R: A Beginner’s Guide" (2023) by Robert N. Stinerock
2	"Research Methods in Education" by Louis Cohen, Lawrence Manion, and Keith Morrison (9th Edition, 2023):
3	"Intellectual Property in the New Technological Age: 2023" by Peter S. Menell, Mark A. Lemley, and Robert P. Merges
4	"Research Handbook on Intellectual Property and Moral Rights" edited by Paul Torremans (2023)

On-Line Materials & Resources (Video experiments / You-tube Videos / Virtual labs / Demo experiments videos, etc…):

1	https://link.springer.com/book/10.1007/978-981-19-5441-2

2	https://onlinecourses.swayam2.ac.in/ntr24_ed08/preview
3	https://ayushnext.ayush.gov.in/index.php/detail/writeUps/national-level-workshop-on-research-methodology-and-ipr
4	https://onlinecourses.nptel.ac.in/noc23_ge36/preview

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2					2	2	2		1
CO2	3	3	3	2				2	2	2		1
CO3	3	3	3	2	1			2	2	2		1
CO4	3	2	3	2	2			2	2	2		1

Note: AAT components must be given to satisfy attainment of PO 8, 9 and 10

Weightage values: 3 – Strongly matched, 2 – moderately matched, 1 – weakly matched, (---) not matched.